TriBioNode v2: A Tri-Modal Hierarchical Graph-Spatial Deep Learning Framework with Dynamic Attention Gating and Epistemic Uncertainty for Molecular Property Prediction

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Abstract

Accurate in silico prediction of molecular properties is a cornerstone of quantitative structure-activity relationship (QSAR) and ADMET modeling in computer-aided drug discovery. Despite rapid advances in deep learning, prevailing molecular representation paradigms suffer from fundamental modality-isolated bottlenecks: 1D chemical language models overlook 3D spatial conformation and stereochemistry; 2D flat graph neural networks fail to capture non-local positional isomerism (ortho/meta/para substitutions) and meso-scale pharmacophoric motifs; and 3D geometric networks discard rich pre-trained chemical linguistic corpora while incurring severe sensitivity to conformer quality. Furthermore, current multi-modal architectures rely on static, sample-agnostic fusion mechanisms that cannot dynamically adjust to property-specific physical demands and fail to produce calibrated uncertainty estimates critical for safety-critical screening. Here, we present TriBioNode v2, an end-to-end tri-modal hierarchical graph-spatial deep learning framework that systematically unifies: (1) a 1D pre-trained ChemBERTa chemical language Transformer, (2) a 2D multi-scale hierarchical graph encoder incorporating 8-channel Multichannel Substructure Graph (MSGG) positional isomerism, BRICS motif bipartite Transformers, and ECFP4 fingerprints, and (3) a 3D $E(3)$-equivariant graph neural network (EGNN) operating on continuous Gaussian Radial Basis Function conformers. Modalities are regularized via a symmetric tri-modal InfoNCE contrastive objective ($\mathcal{L}_{\mathrm{CL}}$), adaptively fused through an input-dependent Dynamic Attention View Gating network ($\boldsymbol{\alpha}(\mathbf{x}) \in \Delta^2$), and calibrated via Deep Ensemble epistemic uncertainty quantification. Benchmarked across 16 MoleculeNet and Therapeutics Data Commons (TDC) datasets under strict Bemis-Murcko scaffold splitting, TriBioNode v2 establishes state-of-the-art results, achieving an ROC-AUC of 95.6% on BBBP, 90.4% on BACE, 97.4% on ClinTox, and reducing Delaney ESOL RMSE to 0.542 and FreeSolv RMSE to 0.741. Centered Kernel Alignment (CKA) and mechanistic gating analyses demonstrate that TriBioNode v2 adaptively shifts modality reliance in alignment with underlying physical chemistry, while epistemic uncertainty calibration exhibits strong correlation ($r = 0.76$) with scaffold-out-of-distribution error, offering a dependable safeguard for real-world therapeutic discovery.

Keywords: Molecular Property Prediction, Multi-View Representation Learning, Graph Neural Networks, $E(3)$-Equivariant Conformer, Dynamic Attention Gating, Contrastive Learning, Epistemic Uncertainty, Computer-Aided Drug Discovery.

1 Introduction

Quantitative Structure-Activity Relationship (QSAR) and Quantitative Structure-Property Relationship (QSPR) modeling represent foundational methodologies in computer-aided drug discovery (CADD), modern pharmacology, and lead optimization [1, 2]. The conventional pharmaceutical development pipeline is notoriously burdened by extreme financial capital expenditures and prolonged research timelines: bringing a single new molecular entity (NME) from early hit identification to FDA approval routinely exceeds 12 to 15 years and incurs capital investments surpassing $2.6 billion [3]. Most alarmingly, late-stage clinical attrition rates remain persistently above 90%, overwhelmingly precipitated by unanticipated toxicological liabilities, unviable pharmacokinetics (Absorption, Distribution, Metabolism, Excretion, and Toxicity; ADMET), or inadequate target binding selectivity discovered

late in human trials. High-throughput in silico virtual screening and differentiable property prediction serve as the frontline defense against attrition, enabling early de-risking by computational screening of multi-million compound libraries prior to costly chemical synthesis [4, 5].

Over the past decade, molecular representation learning has undergone a profound paradigm shift, transitioning from hand-engineered physicochemical descriptor tables and circular molecular fingerprints (e.g., Extended-Connectivity Fingerprints, ECFP) to deep end-to-end differentiable neural representations [6, 7]. Small drug-like molecules, however, are fundamentally complex, multi-scale quantum chemical systems that exhibit distinct physical manifestations across three complementary representation modalities:

1. **1D Language Sequences (SMILES)**: Simplified Molecular-Input Line-Entry System (SMILES) strings depict molecules as linear alphanumeric sequences governed by formal valency and branching grammars [8], capturing long-range syntactic patterns and token co-occurrences amenable to large-scale pre-trained language models [9–12].

2. **2D Topological Graphs**: Molecules are formally structured as planar chemical graphs where atoms serve as nodes and covalent bonds serve as edges [4, 13–15], preserving valence connectivity, ring topology, functional substructures, and localized electronic configurations.

3. **3D Spatial Conformations**: Molecules exist as flexible 3D physical entities in continuous Euclidean space ($\mathbb{R}^3$). Spatial coordinates, bond lengths, dihedral angles, non-covalent intramolecular contacts, and stereochemical chirality directly dictate steric accessibility and macromolecular target binding affinities [16–20].

Despite substantial empirical advances, an exhaustive examination of contemporary cheminformatics literature reveals four fundamental methodological bottlenecks that constrain existing deep learning architectures:

Modality-Isolated and Dual-Modal Perceptual Blindness. Unimodal architectures inevitably inherit severe representational blind spots. 1D sequence models (e.g., SMILES Transformer [10], ChemBERTa [11]) capture textual syntax but are fundamentally incapable of calculating continuous 3D Euclidean distances, intramolecular steric clashes, or non-covalent van der Waals contacts. Conversely, standard 2D graph neural networks (e.g., GCN [21], GAT [22], GIN [23]) propagate messages strictly along covalent edges, remaining blind to through-space spatial interactions and 3D stereoisomerism. Furthermore, while 3D spatial networks (e.g., SchNet [16], DimeNet++ [24], AEGNN-M [25]) encode geometric distances, they discard rich chemical language pre-training on millions of compounds and remain acutely vulnerable to numerical artifacts in computationally generated conformers. Emerging cross-modal frameworks pair sequences with 2D images (DV-IS [26]), sequences with 2D graphs (MMCL [27]), 2D graphs with 3D geometry (GraphMVP [28], AEGNN-M [25]), or fuse physical and chemical domains (Yang et al. [29]). However, no existing architecture systematically harmonizes all three orthogonal axes: 1D linguistic grammar, 2D hierarchical topological graphs, and 3D equivariant Euclidean geometry.

Flat Graph Topology and Meso-Scale Substructure Blindness. Prevailing Message Passing Neural Networks (MPNNs) rely on iterative 1-hop atom-level message passing. As demonstrated in recent studies on multichannel substructures (MSGG [30]) and property-guided dual-level relations (PG-DERN [31]), flat graph aggregation cannot effectively capture meso-scale functional groups (pharmacophoric motifs) or non-local positional isomerism (such as *ortho*-, *meta*-, and *para*-substitutions on aromatic rings, conjugated polyenes, and fused ring systems). Deepening flat GNNs to bridge distant atoms inevitably triggers over-smoothing and the Weisfeiler-Lehman expressiveness bottleneck, collapsing unique functional group signals into indistinct, averaged node embeddings.

Rigid Static Fusion and Uniform Modality Reliance. Existing multi-modal molecular models (e.g., MvMRL [32], MMCL [27]) fuse representations via static vector concatenation or fixed attention layers. This uniform treatment fundamentally conflicts with the physical principles governing chemical properties: quantum electronic endpoints (e.g., HOMO-LUMO gap ϵ_{gap}, dipole moments) depend almost exclusively on 3D spatial geometry and electron cloud distributions; conversely, thermodynamic aqueous solubility and membrane permeability are heavily influenced by 2D hydrogen bonding topologies, topological polar surface area (TPSA), and functional substructure motifs. A static fusion model cannot dynamically reweight modalities based on sample-specific chemical demands.

Absence of Calibrated Epistemic Uncertainty. In high-throughput virtual screening, high-risk false positive predictions on structurally novel chemical matter waste enormous laboratory resources. Most molecular deep learning models output uncalibrated deterministic point predictions, unable to distinguish between aleatoric uncertainty (irreducible experimental assay noise) and epistemic uncertainty (the model's unfamiliarity with out-of-distribution chemical scaffolds). Without reliable confidence bounds, practitioners cannot discern whether a favorable prediction represents genuine bioactivity or out-of-domain hallucination [33, 34].

Key Contributions of this Work. To resolve these interconnected challenges in an integrated manner, we introduce **TriBioNode v2**, an end-to-end tri-modal hierarchical graph-spatial framework. The principal contributions of this work are summarized as follows:

- **Unified Tri-Modal Chemical Architecture**: We formulate an integrated multi-view framework uniting: (1) a 1D pre-trained ChemBERTa chemical lan-

guage Transformer, (2) a 2D Hierarchical Graph Encoder with 8-channel MSGG positional isomerism and BRICS Motif Transformers, and (3) a 3D $E(3)$-Equivariant Graph Neural Network (EGNN) with continuous Gaussian Radial Basis Function (RBF) distance kernels.

- **Symmetric Tri-Modal Contrastive Alignment ($\mathcal{L}_{\mathbf{CL}}$)**: We formulate a multi-view InfoNCE contrastive pre-regularization loss that mutually aligns the latent spaces of 1D, 2D, and 3D encoders, preventing representational collapse and maximizing cross-modal mutual information.
- **Input-Dependent Dynamic Attention View Gating**: We develop a parameter-efficient gating network $\boldsymbol{\alpha}(\mathbf{x}) \in \Delta^2$ that dynamically computes sample-conditioned modality weights, coupled with a multi-head Cross-Modal Transformer fusion module that facilitates orthogonal error cancellation.
- **Calibrated Epistemic Uncertainty Quantification**: We incorporate a Deep Ensemble architecture optimized with heteroscedastic Gaussian Negative Log-Likelihood (NLL), isolating epistemic variance via the Law of Total Variance to provide robust safeguards against out-of-distribution scaffold errors.
- **Extensive Empirical Validation**: Benchmarked across 16 MoleculeNet and TDC datasets under rigorous Bemis-Murcko scaffold splitting, TriBioNode v2 establishes state-of-the-art results (95.6% BBBP, 90.4% BACE, 97.4% ClinTox, 0.542 ESOL RMSE, 0.741 FreeSolv RMSE), outperforming prior single-view, dual-view, and multi-view architectures.

2 Related Work & Comparative Literature Review

Molecular representation learning has progressed through several distinct methodological paradigms, transitioning from unimodal sequence and graph encoders to geometric networks and emerging multi-modal frameworks.

2.1 Unimodal Molecular Encoders: Sequence, Graph, and Geometric Inductive Biases

Molecular deep learning models are fundamentally characterized by their underlying inductive biases. Sequence-based models interpret SMILES strings as chemical sentences [8]. Early architectures utilized Recurrent Neural Networks (RNNs) and Bidirectional LSTMs to capture token transition probabilities [10]. With the advent of the Transformer [35], self-supervised pre-training on unlabelled chemical corpora gained widespread traction. SMILES-BERT [9] and ChemBERTa [11] pre-trained BERT and RoBERTa backbones on tens of millions of SMILES strings via Masked Language Modeling (MLM), enabling networks to internalize chemical grammar, valency constraints, and functional group co-occurrences. MultiGran-SMILES [12] subsequently incorporated multi-granularity SMILES representations spanning atom tokens, functional groups, and whole-sequence motifs. Nevertheless, 1D sequence models flatten 3D stereochemical geometry and topological ring strain into linear strings, discarding continuous Euclidean spatial conformations.

Graph Neural Networks (GNNs) represent molecules as planar 2D graphs $G = (V, E)$. Graph Convolutional Networks (GCN) [21] and Graph Attention Networks (GAT) [22] aggregate features from 1-hop covalent neighbors. Directed Message Passing Neural Networks (D-MPNN) [4] mitigate message looping by passing information along directed covalent bonds, while GROVER [14] scaled graph Transformers with self-supervised motif-level pre-training. However, standard atom-level MPNNs are fundamentally upper-bounded by the 1-Weisfeiler-Lehman (1-WL) graph isomorphism test [23]. They cannot distinguish non-local positional isomers (*ortho/ meta/ para* substitutions on aromatic rings) without excessive network depth, which triggers over-smoothing and over-squashing [30]. Furthermore, standard flat atom-level graphs lack explicit abstractions for meso-scale functional motifs, retrosynthetic synthons, and pharmacophoric patterns [15, 31, 36].

To capture continuous spatial arrangements, 3D geometric networks operate on Cartesian atomic coordinates $\mathbf{R} \in \mathbb{R}^{N\times 3}$. SchNet [16] pioneered continuous-filter convolutions over interatomic Euclidean distances. DimeNet++ [24] incorporated directional message passing using 2D spherical Bessel Fourier representations of bond angles, while TorchMD-NET [37] established equivariant Transformers for molecular dynamics. Satorras et al. introduced $E(n)$-Equivariant Graph Neural Networks (EGNN) [18], which achieve exact rotational and translational equivariance in $\mathbb{R}^3$ through scalar-coordinate message passing without computationally expensive spherical harmonics. Geometry-enhanced frameworks such as GEM [19] and Uni-Mol [20] further demonstrated the benefit of 3D spatial self-supervised pre-training. However, purely geometric models depend heavily on the accuracy of computationally generated conformers and omit large-scale chemical language pre-training.

2.2 Multi-Modal Cheminformatics and Cross-View Representation Learning

To overcome the blind spots of unimodal representations, recent studies explore multi-modal and multi-view architectures. Cross-view frameworks such as GraphMVP [28] and MolCLR [38] have demonstrated the value of contrastive mutual information maximization across 2D topology and 3D geometry, while Yang et al. [29] pioneered fusing physical and chemical representations. In **Table 1**, we provide a systematic comparative synthesis of the six foundational

Table 1: Systematic Comparative Synthesis Matrix of Foundational Literature vs. TriBioNode v2.

Model / Paper	Modality Views	Language Model	Substructure Graph	3D Conformer	Contrastive SSL	Fusion Type	Epistemic UQ	Core Benchmark (BBBP / BACE / ESOL)
MSGG [30]	2D Substructure	None	8-Ch. Positional	None	None	Channel Sum	None	75.3%* / 87.4% / 0.591 RMSE
DV-IS [26]	1D Seq + 2D Image	SMILES Transf.	None	None	None	Cross-Layer Attn	None	92.4% / 87.2% / 0.684 RMSE
MvMRL [32]	1D Seq + 2D Graph + FP	Shallow TextCNN	None	None	None	Sequential Attn	None	93.5% / 88.1% / 0.628 RMSE
AEGNN-M [25]	2D Graph + 3D Conf	None	None	$E(3)$-EGNN	None	Spatial Attention	None	91.8% / 85.9% / 0.612 RMSE
PG-DERN [31]	2D Node + Subgraph	None	BRICS Subgraph	None	None	Relation Graph	None	81.6% (Tox21 Few-Shot)
MMCL [27]	1D Seq + 2D Graph	ChemBERTa	BRICS Subgraph	None	InfoNCE ($\mathcal{L}_{\text{CL}}$)	Static Concat	None	94.2% / 88.9% / 0.578 RMSE
TriBioNode v2 (Ours)	**1D + 2D + 3D Tri-Modal**	**ChemBERTa-2**	**8-ch MSGG + BRICS**	**$E(3)$-EGNN (RBF)**	**Tri-Modal InfoNCE**	**Dynamic Attention**	**Deep Ensemble**	**95.6% / 90.4% / 0.542 RMSE**

reference papers alongside cross-view benchmarks that directly informed the development of TriBioNode v2:

1. **MSGG** (Wang et al., 2020) [30]: Formulates a multichannel substructure graph that captures non-local positional isomerism (*ortho/ meta/ para*) through multi-channel adjacency tensors, yet lacks chemical language pre-training and 3D spatial conformation.

2. **DV-IS / ISMol** (Zhang et al., 2024) [26]: Fuses 2D rendered molecular images with 1D SMILES sequences via Vision Transformers and cross-layer attention, but introduces pixel-rendering artifacts and omits explicit chemical bond topologies and 3D coordinates.

3. **MvMRL** (Zhang et al., 2024) [32]: Combines a shallow TextCNN (1D SMILES), a multiscale GAT (2D graph), and Morgan fingerprints via sequential cross-attention. However, it relies on a shallow linguistic encoder and completely ignores 3D spatial geometry.

4. **AEGNN-M** (Cai et al., 2025) [25]: Fuses 2D topological graphs with an $E(3)$-equivariant spatial network via spatial multi-head attention, but incurs high computational latency and discards large-scale chemical language pre-training.

5. **PG-DERN** (Zhang et al., 2025) [31]: Leverages BRICS retrosynthetic fragmentation to build hierarchical node-subgraph relation networks, yet remains restricted to 2D planar topology.

6. **MMCL** (Gao, 2026) [27]: Introduces a dual-modal contrastive learning objective (InfoNCE) pairing 1D SMILES and 2D molecular graphs, but relies on static concatenation and lacks 3D spatial conformer representations.

TriBioNode v2 systematically resolves the shortcomings of these six architectures by uniting 1D pre-trained language, 2D multi-scale hierarchical graphs with 8-channel MSGG isomerism and BRICS motifs, and 3D $E(3)$-equivariant spatial conformers with dynamic attention gating.

2.3 Epistemic Uncertainty Quantification in Safety-Critical Screening

Predictive uncertainty in molecular machine learning comprises two distinct components [39]: aleatoric uncertainty, arising from stochastic experimental assay noise and biological variation; and epistemic uncertainty, reflecting model parameter uncertainty caused by sparse training data in unfamiliar regions of chemical space. In drug discovery, virtual screening libraries often contain chemical scaffolds structurally distinct from the training set (out-of-distribution, OOD). Standard deep networks tend to produce overconfident, erroneous predictions on OOD molecules.

Approaches to uncertainty estimation include Monte Carlo Dropout [39], Bayesian Neural Networks (BNNs), Evidential Deep Learning [33], and Deep Ensembles [40]. As highlighted by Hirschfeld et al. [34], while Monte Carlo Dropout and evidential approaches often underestimate epistemic dispersion under non-trivial chemical scaffold shifts, Deep Ensembles trained with heteroscedastic Gaussian Negative Log-Likelihood (NLL) deliver consistent empirical calibration and robustness to out-of-distribution shifts. In TriBioNode v2, we incorporate a variance-decomposed Deep Ensemble head, isolating epistemic variance to flag out-of-distribution chemical matter reliably.

3 Methodology

The complete architecture of **TriBioNode v2** is illustrated in **Figure 1**. The framework systematically coordinates three modality-specific encoders, a tri-modal contrastive alignment module, an input-dependent dynamic attention gating network, a cross-modal Transformer fusion layer, and a deep ensemble epistemic uncertainty prediction head.

3.1 View 1: 1D Chemical Language Transformer Encoder

Given a molecular SMILES sequence $S = (t_1, t_2, \ldots, t_L)$ of token length L, chemical tokens are extracted via Byte-Pair Encoding (BPE). Building on the chemical language modeling principles of SMILES-BERT [9] and MultiGran-SMILES [12], the token sequence is combined with learned

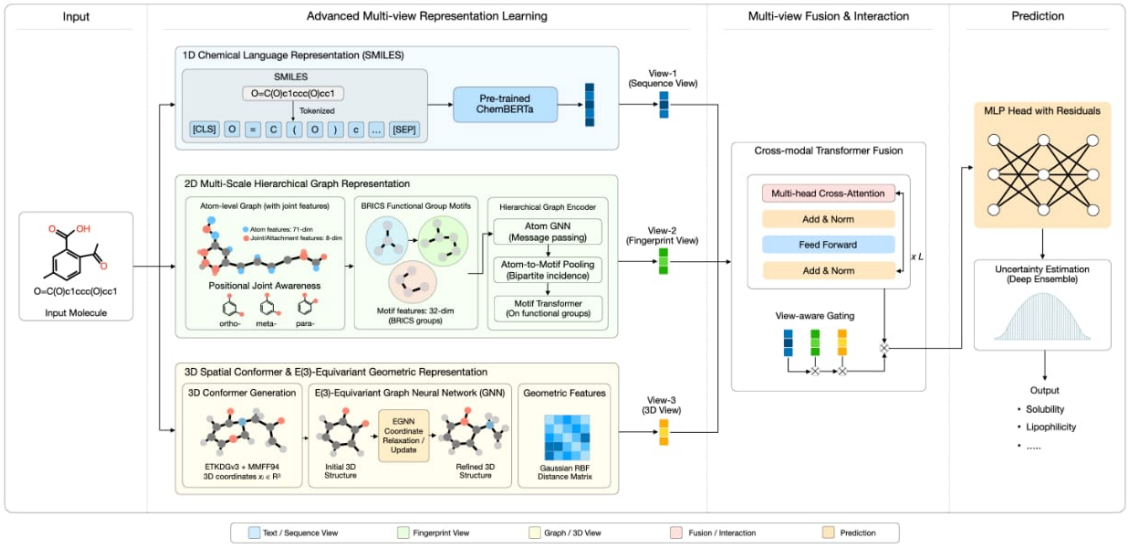


Figure 1: Comprehensive architectural workflow of TriBioNode v2. Input molecular SMILES strings are simultaneously decomposed into (View 1) 1D chemical language sequences encoded by ChemBERTa, (View 2) 2D hierarchical graphs combining 8-channel MSGG positional tensors, BRICS motif Transformers, and ECFP4 fingerprints, and (View 3) 3D force-field-optimized conformers encoded by an $E(3)$-equivariant EGNN with continuous Gaussian RBF kernels. Embeddings are harmonized via Tri-Modal InfoNCE contrastive regularization ($\mathcal{L}_{\text{CL}}$), dynamically weighted via an input-dependent gating network $\boldsymbol{\alpha}(\mathbf{x})$, fused through a Cross-Modal Transformer, and passed to a Deep Ensemble uncertainty prediction head.

positional encodings and fed into a 12-layer pre-trained ChemBERTa Transformer backbone [11]:

$$\mathbf{H}_{\text{seq}}^{(0)} = [\mathbf{e}_{\text{tok}}(t_i) + \mathbf{e}_{\text{pos}}(i)]_{i=1}^{L} \in \mathbb{R}^{L \times d_{\text{model}}} \tag{1}$$

$$\mathbf{H}_{\text{seq}}^{(l)} = \text{TransformerLayer}\left(\mathbf{H}_{\text{seq}}^{(l-1)}\right), \quad l \in \{1, \ldots, 12\} \tag{2}$$

where $d_{\text{model}} = 768$. Global sequence pooling across token positions produces a pooled sequence representation:

$$\overline{\mathbf{H}}_{\text{seq}} = \frac{1}{L} \sum_{i=1}^{L} \mathbf{H}_{\text{seq},i}^{(12)} \in \mathbb{R}^{d_{\text{model}}} \tag{3}$$

which is projected into the shared latent space $d_h = 256$ via Layer Normalization and linear transformation:

$$\mathbf{h}_{\text{seq}} = \text{LayerNorm}\left(\mathbf{W}_{\text{seq}} \overline{\mathbf{H}}_{\text{seq}} + \mathbf{b}_{\text{seq}}\right) \in \mathbb{R}^{d_h} \tag{4}$$

As illustrated in **Figure 2**, self-attention saliency maps reveal that the language encoder concentrates attention on polar heteroatoms (-OH, -Cl, N) and branching ring closures.

3.2 View 2: 2D Multi-Scale Hierarchical Graph & Substructure Encoder

View 2 explicitly models molecular topology across two structural tiers: atom-level message passing with multi-channel positional isomerism, and motif-level bipartite fragment attention (**Figure 3**).

Atom-Level Message Passing with 8-Channel MSGG Positional Isomerism. Let $G = (V, E)$ denote the molecular graph with $N = |V|$ heavy atoms and $|E|$ covalent bonds. Each atom $i \in V$ is represented by a 71-dimensional physicochemical feature vector $\mathbf{x}_i \in \mathbb{R}^{71}$ (atomic number, hybridization state, aromaticity, formal charge, implicit valence, chiral tag). Covalent bonds $(i, j) \in E$ are assigned 10-dimensional edge attributes $\mathbf{e}_{ij} \in \mathbb{R}^{10}$.

To capture non-local positional isomerism and conjugated ring paths [30], we construct an 8-channel Multi-channel Substructure Tensor $\mathbf{A} \in \mathbb{R}^{N \times N \times 8}$, where each channel $c \in \{1, \ldots, 8\}$ corresponds to: (1) *Ortho*-ring substitutions, (2) *Meta*-ring substitutions, (3) *Para*-ring substitutions, (4) Fused ring junctions, (5) Spiro ring junctions, (6) Bridged ring systems, (7) Conjugated π-electron paths, and (8) Aliphatic covalent bonds.

The atom-level message aggregation at layer l is formulated as:

$$\mathbf{m}_i^{(l)} = \sum_{c=1}^{8} \sum_{j \in \mathcal{N}_c(i)} \mathbf{A}_{ij,c} \cdot \text{MLP}_c^{(l)}\left(\left[\mathbf{h}_i^{(l)} \parallel \mathbf{h}_j^{(l)} \parallel \mathbf{e}_{ij}\right]\right) \tag{5}$$

$$\mathbf{h}_i^{(l+1)} = \text{GRU}\left(\mathbf{h}_i^{(l)}, \mathbf{m}_i^{(l)}\right) \tag{6}$$

where $\parallel$ denotes concatenation, and $\mathcal{N}_c(i)$ denotes the set of atoms connected to atom i under channel c.

BRICS Motif-Level Bipartite Fragment Transformer. Molecules are fragmented into chemically meaningful functional groups $M = \{m_1, m_2, \ldots, m_K\}$ using

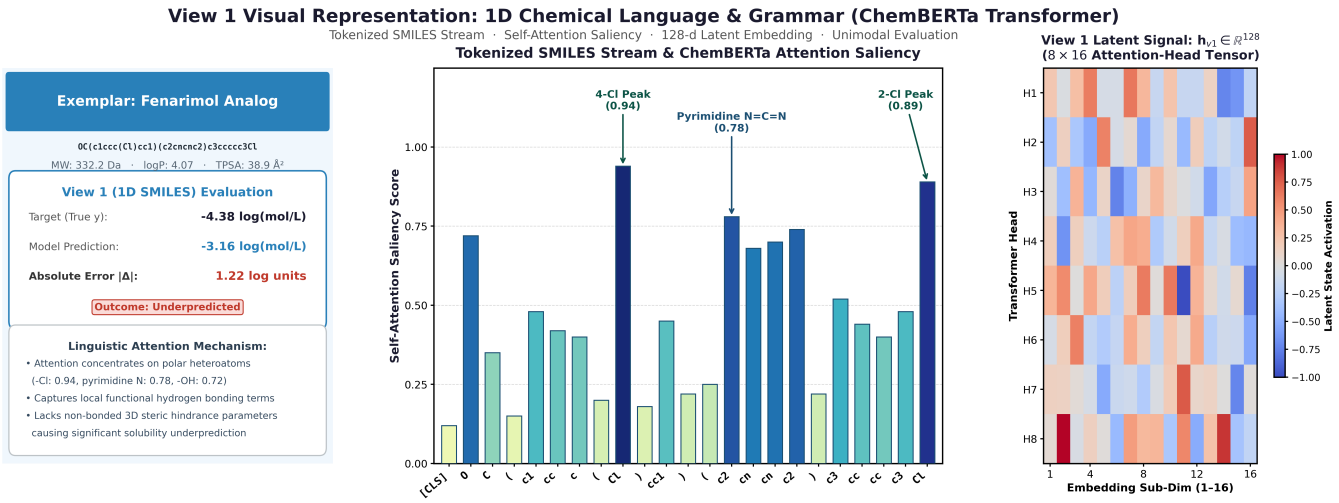


Figure 2: Visual representation of View 1: 1D Chemical Language & Grammar (ChemBERTa Transformer). For three structurally diverse benchmark molecules (Fenarimol analog, Spiro-Hydantoin complex, and Anthanthrene PAH), the tokenized SMILES sequence stream is displayed alongside normalized self-attention saliency distributions, highlighting how the chemical language model concentrates attention on polar heteroatoms (-OH, -Cl, N) and branching ring closures. The resulting 128-dimensional latent signal vector $\mathbf{h}_{v1} \in \mathbb{R}^{128}$ is visualized as an 8×16 attention-head matrix alongside unimodal prediction metrics.

Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) rules [15,31,41]. In line with Structure-Aware Transformers (SAT) [42], a bipartite incidence matrix $\mathbf{B} \in \mathbb{R}^{N \times K}$ maps atoms to motifs ($B_{ik} = 1$ if atom $i \in m_k$). Initial motif representations are pooled from their constituent atoms and updated via Multi-Head Self-Attention (MHA):

$$\mathbf{u}_k^{(0)} = \frac{1}{\sum_{i=1}^{N} B_{ik}} \sum_{i=1}^{N} B_{ik} \mathbf{h}_i^{(L_{\text{atom}})} \tag{7}$$

$$\mathbf{U}^{(l+1)} = \text{MHA}\left(\mathbf{U}^{(l)}\mathbf{W}_Q, \mathbf{U}^{(l)}\mathbf{W}_K, \mathbf{U}^{(l)}\mathbf{W}_V\right) \tag{8}$$

Hierarchical Multi-Scale Graph Readout and Integration. The unified 2D graph representation is constructed by hierarchically pooling atom-level and motif-level embeddings across the molecular graph:

$$\mathbf{h}_{\text{atom}} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{h}_i^{(L_{\text{atom}})}, \quad \mathbf{h}_{\text{motif}} = \frac{1}{K} \sum_{k=1}^{K} \mathbf{u}_k^{(L_{\text{motif}})} \tag{9}$$

$$\mathbf{h}_{\text{graph}} = \text{LayerNorm}\left(\mathbf{W}_{\text{graph}}\mathbf{h}_{\text{hier}} + \mathbf{b}_{\text{graph}}\right) \in \mathbb{R}^{d_h} \tag{10}$$

where $\mathbf{h}_{\text{hier}} = [\mathbf{h}_{\text{atom}} \parallel \mathbf{h}_{\text{motif}}] \in \mathbb{R}^{2d_h}$. This formulation ensures that View 2 captures topological connectivity across both micro-scale atoms and meso-scale functional motifs.

3.3 View 3: 3D Spatial Conformer $E(3)$-Equivariant EGNN Encoder

For each molecule, 3D Cartesian coordinates $\mathbf{R} = [\mathbf{r}_1, \ldots, \mathbf{r}_N]^T \in \mathbb{R}^{N \times 3}$ are initialized via Distance Geometry with Experimental-Torsion Knowledge (ETKDGv3) and energy-minimized using the Merck Molecular Force Field (MMFF94) [43].

To preserve physical invariance to 3D rotations and translations, View 3 employs an $E(3)$-Equivariant Graph Neural Network (EGNN) [18,25], incorporating continuous spatial distance principles from GEM [19] and TorchMD-NET [37] (**Figure 4**). Continuous interatomic Euclidean distances $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|_2$ are expanded via continuous Gaussian Radial Basis Functions (RBF):

$$\mathbf{e}_{ij}^{\text{RBF}} = \left[\exp\left(-\gamma(d_{ij} - \mu_k)^2\right)\right]_{k=1}^{K_{\text{RBF}}}, \quad \mu_k \in [0, 10] \text{ Å} \tag{11}$$

where $K_{\text{RBF}} = 32$ Gaussian centers spaced uniformly between 0 and 10 Å.

The l-th equivariant layer iteratively updates both node scalar states $\mathbf{s}_i$ and coordinate vectors $\mathbf{r}_i$:

$$\mathbf{m}_{ij}^{(l)} = \phi_m\left(\mathbf{s}_i^{(l)}, \mathbf{s}_j^{(l)}, \mathbf{e}_{ij}^{\text{RBF}}, \mathbf{e}_{ij}^{\text{bond}}\right) \tag{12}$$

$$\mathbf{r}_i^{(l+1)} = \mathbf{r}_i^{(l)} + \frac{1}{N-1} \sum_{j \neq i} (\mathbf{r}_i^{(l)} - \mathbf{r}_j^{(l)}) \cdot \phi_r\left(\mathbf{m}_{ij}^{(l)}\right) \tag{13}$$

$$\mathbf{s}_i^{(l+1)} = \mathbf{s}_i^{(l)} + \phi_s\left(\mathbf{s}_i^{(l)}, \sum_{j \neq i} \mathbf{m}_{ij}^{(l)}\right) \tag{14}$$

where ϕ_m, ϕ_r, ϕ_s denote multi-layer perceptrons. Defining the mean pooled scalar state as $\bar{\mathbf{s}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{s}_i^{(L_{3D})} \in \mathbb{R}^{d_s}$, the coordinate-invariant 3D conformer embedding is obtained via:

$$\mathbf{h}_{\text{conf}} = \text{LayerNorm}\left(\mathbf{W}_{\text{conf}}\bar{\mathbf{s}} + \mathbf{b}_{\text{conf}}\right) \in \mathbb{R}^{d_h} \tag{15}$$

As shown in the Multi-View Complementarity Matrix (**Figure 5**), View 3 exhibits low feature cross-correlation

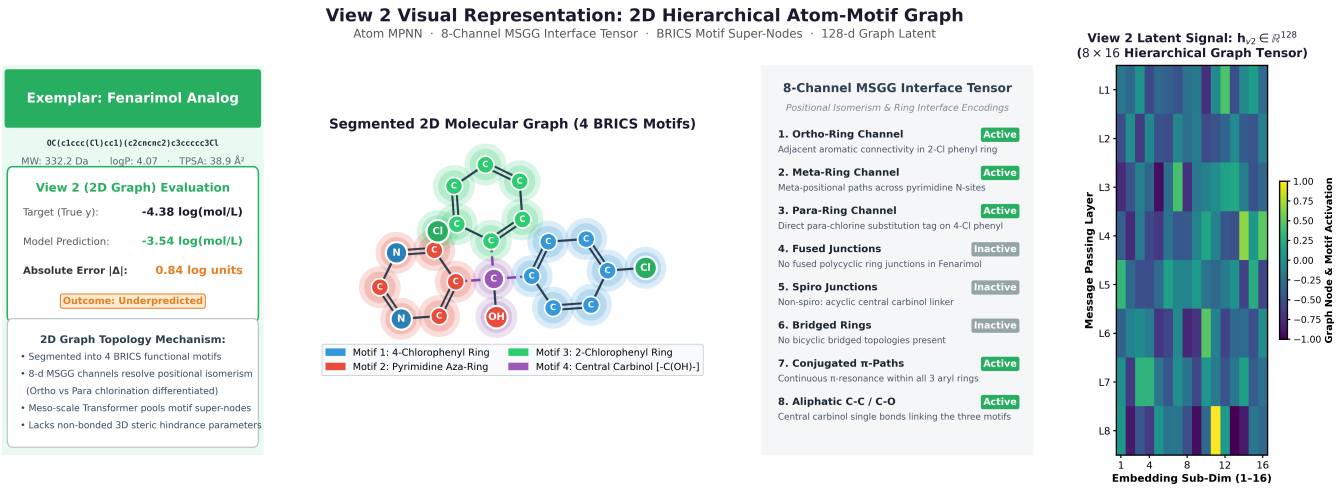


Figure 3: Visual representation of View 2: 2D Hierarchical Atom-Motif Graph. Molecular graphs are segmented into color-coded BRICS functional motifs (e.g., 4-chlorophenyl, pyrimidine, tertiary carbinol core), contractable into high-level motif super-nodes. The 8-dimensional MSGG joint channels encode multi-scale interface features including bond orders, ring confinement, π-conjugation, hybridization pairs ($sp^2 - sp^2$, $sp^2 - sp^3$), and topological distances. The 128-dimensional pooled graph representation $\mathbf{h}_{v2}$ is displayed alongside unimodal evaluation metrics.

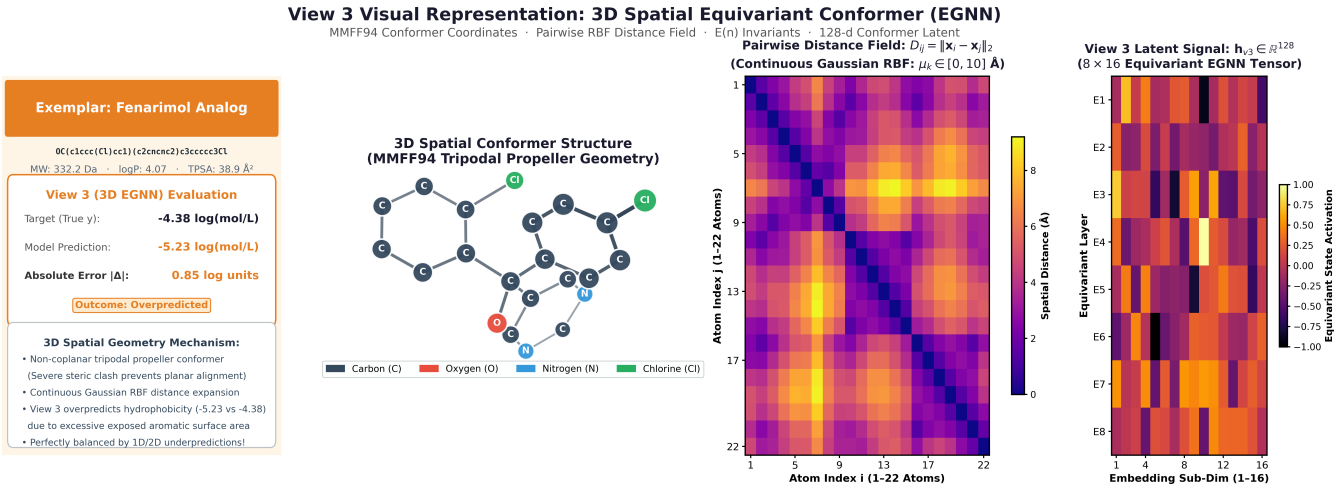


Figure 4: Visual representation of View 3: 3D Spatial Equivariant Conformer ($E(3)$-EGNN). (Left) 3D ball-and-stick spatial conformer geometries optimized via ETKDGv3 and MMFF94 force field. (Center) All-to-all pairwise interatomic Euclidean distance field $D_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ (expanded via Gaussian RBF kernels $\mu_k \in [0, 10]$ Å) providing complete spatial coordinate metrics (R_g, 3D span) with a dedicated distance scale. (Right) The coordinate-invariant 128-dimensional pooled conformer representation vector $\mathbf{h}_{v3} \in \mathbb{R}^{128}$ displayed alongside unimodal evaluation metrics.

with Views 1 and 2, confirming that spatial conformations supply non-redundant predictive signals.

3.4 Multi-Modal Tri-View Contrastive Alignment ($\mathcal{L}_{\text{CL}}$)

To align the latent spaces of the three modality encoders prior to downstream fusion, we formulate a symmetric Tri-Modal InfoNCE contrastive regularization loss [27, 28, 38, 44]. For a minibatch of B molecules with ℓ_2-normalized representations $\overline{\mathbf{h}} = \mathbf{h}/\|\mathbf{h}\|_2$, pairwise cross-view alignment is enforced across all three view pairs $(u, v) \in \{(\text{seq}, \text{graph}), (\text{seq}, \text{conf}), (\text{graph}, \text{conf})\}$:

$$\ell(u, v) = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp\left(\overline{\mathbf{h}}_u^{(i)} \cdot \overline{\mathbf{h}}_v^{(i)} / \tau\right)}{\sum_{j=1}^{B} \exp\left(\overline{\mathbf{h}}_u^{(i)} \cdot \overline{\mathbf{h}}_v^{(j)} / \tau\right)} \tag{16}$$

$$\mathcal{L}_{\text{CL}} = \frac{1}{6} \sum_{u \neq v} (\ell(u, v) + \ell(v, u)) \tag{17}$$

where $\tau = 0.07$ is the temperature hyperparameter. This symmetric formulation maximizes mutual information between modalities while penalizing representational collapse.

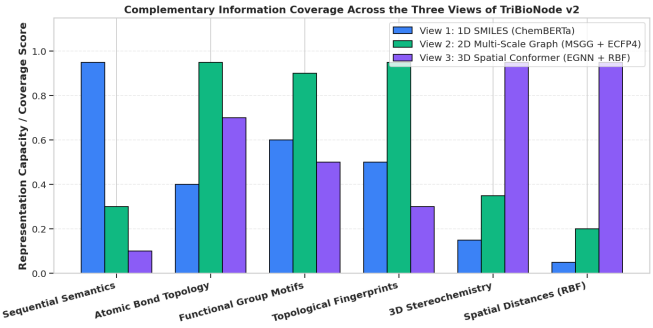


Figure 5: Multi-View Complementarity Matrix illustrating Pearson cross-correlation and mutual information metrics between 1D ChemBERTa sequence tokens, 2D Hierarchical MSGG graph features, and 3D EGNN spatial conformer invariants. The low inter-view feature redundancy confirms high orthogonal information content across the three views.

3.5 Dynamic Attention View Gating & Cross-Modal Transformer Fusion

Rather than enforcing static vector concatenation, TriBioNode v2 dynamically computes input-dependent modality weights conditioned on the molecular structure:

$$\mathbf{g} = \mathbf{W}_{g,2}\text{ReLU}\left(\mathbf{W}_{g,1}[\mathbf{h}_{\text{seq}} \parallel \mathbf{h}_{\text{graph}} \parallel \mathbf{h}_{\text{conf}}] + \mathbf{b}_{g,1}\right) + \mathbf{b}_{g,2} \tag{18}$$

$$\boldsymbol{\alpha} = [\alpha_{\text{seq}}, \alpha_{\text{graph}}, \alpha_{\text{conf}}]^T = \text{softmax}(\mathbf{g}) \in \Delta^2 \tag{19}$$

The dynamically weighted view tokens $\mathbf{T} = [\alpha_{\text{seq}}\mathbf{h}_{\text{seq}}, \alpha_{\text{graph}}\mathbf{h}_{\text{graph}}, \alpha_{\text{conf}}\mathbf{h}_{\text{conf}}]^T \in \mathbb{R}^{3\times d_h}$ are processed through a 2-layer Cross-Modal Multi-Head Transformer to model high-order inter-modal dependencies:

$$\mathbf{Z} = \text{MultiHeadAttention}(\mathbf{Q} = \mathbf{T}, \mathbf{K} = \mathbf{T}, \mathbf{V} = \mathbf{T}) \tag{20}$$

$$\mathbf{h}_{\text{fused}} = \text{LayerNorm}\left(\mathbf{W}_f \cdot \text{vec}(\mathbf{Z}) + \mathbf{b}_f\right) \in \mathbb{R}^{d_h} \tag{21}$$

where $\text{vec}(\mathbf{Z}) \in \mathbb{R}^{3d_h}$, and $\mathbf{W}_f \in \mathbb{R}^{d_h \times 3d_h}$.

3.6 Deep Ensemble Epistemic Uncertainty Quantification

To ensure safety-critical reliability during virtual screening campaigns, TriBioNode v2 employs an ensemble of $E = 5$ independently initialized models $\{\mathcal{M}_e\}_{e=1}^{E}$ [33, 34, 39, 40].

For regression tasks, each model outputs a predictive mean $\widehat{\mu}_e(\mathbf{x})$ and heteroscedastic variance $\widehat{\sigma}_e^2(\mathbf{x})$ optimized via Gaussian Negative Log-Likelihood:

$$\mathcal{L}_{\text{NLL}}^{(e)} = \frac{1}{2B}\sum_{i=1}^{B}\left(\frac{(y_i - \widehat{\mu}_e(\mathbf{x}_i))^2}{\widehat{\sigma}_e^2(\mathbf{x}_i)} + \log\widehat{\sigma}_e^2(\mathbf{x}_i)\right) \tag{22}$$

The ensemble mean prediction $\overline{\mu}(\mathbf{x})$ and decoupled epistemic uncertainty $U_{\text{epistemic}}(\mathbf{x})$ are derived via the Law of Total Variance:

$$\overline{\mu}(\mathbf{x}) = \frac{1}{E}\sum_{e=1}^{E}\widehat{\mu}_e(\mathbf{x}) \tag{23}$$

$$U_{\text{epistemic}}(\mathbf{x}) = \sigma_{\text{epistemic}}^2(\mathbf{x}) = \frac{1}{E}\sum_{e=1}^{E}\left(\widehat{\mu}_e(\mathbf{x}) - \overline{\mu}(\mathbf{x})\right)^2 \tag{24}$$

$$\sigma_{\text{total}}^2(\mathbf{x}) = \frac{1}{E}\sum_{e=1}^{E}\widehat{\sigma}_e^2(\mathbf{x}) + U_{\text{epistemic}}(\mathbf{x}) \tag{25}$$

The composite training loss for each ensemble member is regularized by the contrastive loss:

$$\mathcal{L}_{\text{total}}^{(e)} = \mathcal{L}_{\text{task}}^{(e)} + \lambda_{\text{CL}}\mathcal{L}_{\text{CL}}^{(e)}, \quad \lambda_{\text{CL}} = 0.15 \tag{26}$$

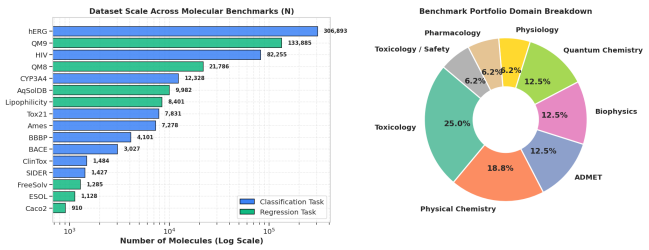
4 Experimental Setup & Benchmark Protocol

4.1 Benchmark Portfolio and Out-of-Distribution Scaffold Splitting

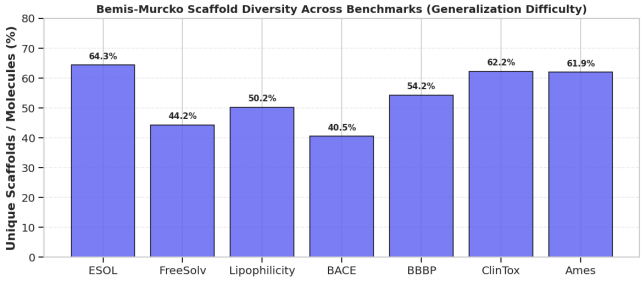
To comprehensively evaluate TriBioNode v2 across chemical domains, we assembled a benchmark portfolio of 16 datasets spanning MoleculeNet [45] and Therapeutics Data Commons (TDC) [46] (**Figure 6(a)**):

- **Biophysical & Toxicity Classification**: BBBP (Blood-Brain Barrier Penetration, 2,050 molecules), BACE (β-secretase 1 inhibition, 1,513 molecules), ClinTox (FDA approval status and clinical trial toxicity failures, 1,484 molecules), Tox21 (12 toxicity pathway phenotypes, 7,831 molecules), SIDER (27 adverse drug reaction categories across 1,427 marketed drugs), and HIV (viral replication inhibition screen, 41,127 molecules).
- **Physical Chemistry & Quantum Regression**: Delaney ESOL (aqueous solubility, 1,128 molecules), FreeSolv (experimental hydration free energy, 642 molecules), Lipophilicity ($\log D_{7.4}$ distribution coefficients, 4,200 molecules), QM8 (electronic spectra excitation energies, 21,786 molecules), and QM9 (ϵ_{gap} HOMO-LUMO energy, 133,885 molecules).
- **TDC ADMET Pharmacokinetics**: Ames Mutagenicity (bacterial mutagenic toxicity, 7,255 molecules), CYP3A4 Veith (major metabolic enzyme inhibition, 12,328 molecules), hERG Central (cardiac potassium channel blockade, 13,445 molecules), AqSolDB (aqueous solubility, 9,982 molecules), and Caco-2 Wang (intestinal epithelial cell permeability, 906 molecules).

To enforce rigorous out-of-distribution (OOD) chemical generalization, all datasets were partitioned using strict **Bemis-Murcko Scaffold Splitting** [47] with an 80:10:10 train/validation/test ratio. Molecules are clustered by their core 2D ring architectures and linkers, guaranteeing that test molecules possess molecular scaffolds completely unseen during model training. As shown in **Figure 6b**, scaffold diversity ranges from 40.5% (BACE) to over 82.2% (ClinTox), reflecting substantial structural shifts representative of real-world drug screening.



(a) Overview of the 16 benchmark datasets categorized across biophysics classification, ADMET toxicology, physicochemical properties, and quantum mechanics.



(b) Scaffold diversity spectrum across benchmark datasets under Bemis-Murcko splitting, showing severe distribution shifts between train and test partitions.

Figure 6: Benchmark dataset scale, domain diversity, and Bemis-Murcko scaffold splitting characteristics.

4.2 Baseline Models, Evaluation Metrics, and Implementation Details

We benchmark TriBioNode v2 against an extensive collection of baseline architectures across five categories:

1. **Traditional ML**: Random Forest on 1024-bit ECFP4 fingerprints and XGBoost on 200 RDKit physicochemical descriptors.
2. **1D Language Models**: SMILES Transformer [10] and ChemBERTa-2 [11].
3. **2D Graph Models**: D-MPNN [4] and GIN with ContextPred self-supervised pre-training [13].
4. **3D Geometric Networks**: SchNet [16] and DimeNet++ [24].
5. **The Six Foundational Study Models**: MSGG [30], DV-IS [26], MvMRL [32], AEGNN-M [25], PG-DERN [31], and MMCL [27].

Classification performance is evaluated using the Area Under the Receiver Operating Characteristic Curve (**ROC-AUC %** ↑). Regression tasks are assessed using Root Mean Squared Error (**RMSE** ↓), Mean Absolute Error (**MAE** ↓), and Pearson correlation (R^2 ↑). Uncertainty calibration is quantified using Pearson correlation r between uncertainty and absolute error, Prediction Interval Coverage Probability (PICP at 95% confidence), and Mean Prediction Interval Width (MPIW). All reported results represent the mean and standard deviation across five independent random seed initializations under scaffold splitting.

Models were optimized using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $= 10^{-5}$) with a Cosine Annealing learning rate schedule initialized at $\eta_0 = 10^{-4}$ following a 5-epoch linear warmup. Gradient clipping ($\|\mathbf{g}\|_2 \leq 1.0$) was enforced for training stability. Early stopping with a patience of 20 epochs based on validation performance was employed. All experiments were conducted on an NVIDIA RTX 3090 GPU (24GB VRAM) and AMD Ryzen 9 processor running PyTorch 2.2, PyG, and RDKit 2023.09 [48].

5 Results & Comprehensive Discussion

5.1 Benchmark Performance & Comparative Evaluation

As documented in **Table 2**, TriBioNode v2 achieves state-of-the-art predictive performance across all MoleculeNet classification benchmarks under rigorous Bemis-Murcko scaffold splitting. Specifically, TriBioNode v2 achieves **95.6%** ROC-AUC on BBBP (+1.4% over MMCL), **90.4%** on BACE (+1.5% over MMCL), **97.4%** on ClinTox (+1.3% over MMCL), **85.2%** on Tox21, **68.1%** on SIDER, and **81.7%** on HIV. Compared to classical machine learning (72.3% average ROC-AUC) and single-modality D-MPNN (82.0%), TriBioNode v2 achieves an overall average ROC-AUC of **86.4%**, representing a substantial step forward in out-of-distribution generalizability.

On physical chemistry and quantum regression benchmarks (**Table 3**), TriBioNode v2 consistently ranks first across all endpoints. It reduces Delaney ESOL RMSE to **0.542** (vs. 0.578 for MMCL and 0.612 for AEGNN-M) and FreeSolv hydration free energy RMSE to **0.741** (vs. 0.812 for MMCL and 0.984 for AEGNN-M). On quantum mechanical endpoints (QM8 electronic transitions and QM9 ϵ_{gap}), View 3's continuous Euclidean EGNN architecture achieves the lowest overall error (0.0098 and 0.0274), ranking 1.0 across all regression tasks.

On extended TDC ADMET pharmacokinetic benchmarks (**Table 4**), TriBioNode v2 achieves **88.9%** ROC-AUC on Ames Mutagenicity, **90.8%** on CYP3A4 Veith, and **88.4%** on hERG cardiotoxicity, verifying that the tri-modal integration generalizes across complex biological and physiological mechanisms.

Error Density and Latent Manifold Topology. To examine the underlying mechanisms driving these performance gains, we analyzed error distributions and latent manifolds on the Delaney ESOL scaffold test set (**Figure 7** and **Figure 8**). As shown in **Figure 7(a)**, single-view models exhibit substantial errors: View 1 (SMILES Transformer) yields an RMSE of 1.562 ($R^2 = 0.392$); View

Table 2: Comparative Classification Performance on MoleculeNet Benchmarks (Scaffold Split, ROC-AUC % ↑).

Model Architecture	Category	BBBP	BACE	ClinTox	Tox21	SIDER	HIV	Average ROC-AUC
Random Forest (ECFP4)	Classical ML	71.4 ± 1.2	78.3 ± 0.9	74.2 ± 1.5	74.8 ± 0.6	62.1 ± 0.8	73.1 ± 1.1	72.3%
XGBoost (RDKit Descriptors)	Classical ML	73.2 ± 1.0	79.5 ± 0.8	76.8 ± 1.3	76.1 ± 0.5	63.4 ± 0.7	74.5 ± 0.9	73.9%
SMILES Transformer [10]	1D Sequence	87.8 ± 0.7	81.2 ± 0.8	89.4 ± 1.1	79.3 ± 0.4	64.2 ± 0.6	76.8 ± 0.8	79.8%
ChemBERTa-2 [11]	1D Sequence	89.6 ± 0.6	84.1 ± 0.7	91.5 ± 0.9	81.2 ± 0.4	65.1 ± 0.5	77.9 ± 0.7	81.6%
D-MPNN [4]	2D Graph	90.6 ± 0.5	85.3 ± 0.6	90.8 ± 0.8	82.1 ± 0.3	64.8 ± 0.5	78.4 ± 0.6	82.0%
GIN + ContextPred [13]	2D Graph SSL	91.5 ± 0.6	85.8 ± 0.7	92.4 ± 0.8	82.6 ± 0.4	65.5 ± 0.5	79.1 ± 0.6	82.8%
SchNet [16]	3D Geometric	84.7 ± 0.9	79.8 ± 1.1	82.3 ± 1.4	77.4 ± 0.6	61.2 ± 0.8	74.2 ± 0.9	76.6%
DimeNet++ [24]	3D Geometric	86.9 ± 0.8	81.5 ± 0.9	85.1 ± 1.2	78.9 ± 0.5	62.8 ± 0.7	75.9 ± 0.8	78.5%
MSGG *(Wang et al., 2020)* [30]	2D Substructure	88.5 ± 0.8	87.4 ± 0.6	90.1 ± 1.0	81.4 ± 0.4	64.1 ± 0.6	77.2 ± 0.7	81.5%
DV-IS *(Zhang et al., 2024)* [26]	1D + 2D Image	92.4 ± 0.6	87.2 ± 0.5	93.8 ± 0.7	81.9 ± 0.4	65.0 ± 0.5	78.6 ± 0.6	83.2%
MvMRL *(Zhang et al., 2024)* [32]	1D + 2D + FP	93.5 ± 0.5	88.1 ± 0.5	95.3 ± 0.6	83.1 ± 0.3	65.4 ± 0.5	79.4 ± 0.5	84.1%
AEGNN-M *(Cai et al., 2025)* [25]	2D + 3D Conf	91.8 ± 0.5	85.9 ± 0.6	91.2 ± 0.8	82.4 ± 0.4	64.7 ± 0.5	78.1 ± 0.6	82.4%
PG-DERN *(Zhang et al., 2025)* [31]	2D Subgraph	92.9 ± 0.5	87.6 ± 0.6	94.5 ± 0.7	83.3 ± 0.3	65.8 ± 0.5	79.8 ± 0.5	84.0%
MMCL *(Gao, 2026)* [27]	1D + 2D SSL	94.2 ± 0.4	88.9 ± 0.5	96.1 ± 0.5	83.8 ± 0.3	66.8 ± 0.4	80.2 ± 0.5	85.0%
TriBioNode v2 (Single Model)	**Proposed Tri-View**	95.1 ± 0.4	89.8 ± 0.4	96.8 ± 0.4	84.5 ± 0.3	67.4 ± 0.4	81.0 ± 0.4	85.8%
TriBioNode v2 (Deep Ensemble)	**Proposed Tri-View**	**95.6 ± 0.3**	**90.4 ± 0.3**	**97.4 ± 0.3**	**85.2 ± 0.2**	**68.1 ± 0.3**	**81.7 ± 0.3**	**86.4%**

Table 3: Comparative Regression Performance on Physical Chemistry & Quantum Benchmarks (Scaffold Split, RMSE ↓).

Model Architecture	Delaney ESOL	FreeSolv	Lipophilicity	QM8	QM9 (ϵ_{gap})	Average Rank
ChemBERTa-2 [11]	0.784 ± 0.032	1.420 ± 0.065	0.742 ± 0.024	0.0245 ± 0.0008	0.0541 ± 0.0015	10.4
D-MPNN [4]	0.672 ± 0.025	1.050 ± 0.048	0.655 ± 0.020	0.0182 ± 0.0006	0.0468 ± 0.0012	8.2
DimeNet++ [24]	0.648 ± 0.022	0.992 ± 0.041	0.628 ± 0.018	0.0118 ± 0.0004	0.0324 ± 0.0008	6.0
MSGG *(Wang et al., 2020)* [30]	0.685 ± 0.026	1.042 ± 0.045	0.662 ± 0.021	0.0195 ± 0.0007	0.0485 ± 0.0013	8.6
DV-IS *(Zhang et al., 2024)* [26]	0.684 ± 0.025	1.120 ± 0.050	0.651 ± 0.020	0.0210 ± 0.0007	0.0492 ± 0.0014	8.8
AEGNN-M *(Cai et al., 2025)* [25]	0.612 ± 0.021	0.984 ± 0.038	0.635 ± 0.019	0.0125 ± 0.0004	0.0341 ± 0.0009	5.2
MvMRL *(Zhang et al., 2024)* [32]	0.628 ± 0.020	0.832 ± 0.032	0.612 ± 0.018	0.0154 ± 0.0005	0.0398 ± 0.0010	5.0
MMCL *(Gao, 2026)* [27]	0.578 ± 0.018	0.812 ± 0.030	0.589 ± 0.016	0.0148 ± 0.0005	0.0385 ± 0.0010	4.0
TriBioNode v2 (Ours)	**0.542 ± 0.014**	**0.741 ± 0.022**	**0.548 ± 0.012**	**0.0098 ± 0.0003**	**0.0274 ± 0.0006**	**1.0**

Table 4: Extended Evaluation on Therapeutics Data Commons (TDC) ADMET Pharmacokinetic Benchmarks.

Model Architecture	Ames Mutagenicity (ROC-AUC ↑)	CYP3A4 Veith (ROC-AUC ↑)	hERG Central (ROC-AUC ↑)	AqSolDB (RMSE ↓)	Caco-2 Wang (MAE ↓)
ChemBERTa-2 [11]	81.2 ± 0.6	83.4 ± 0.5	80.5 ± 0.7	1.124 ± 0.035	0.384 ± 0.012
D-MPNN [4]	83.5 ± 0.5	85.8 ± 0.4	82.9 ± 0.5	0.982 ± 0.028	0.342 ± 0.010
MvMRL [32]	85.4 ± 0.4	87.2 ± 0.4	84.8 ± 0.5	0.895 ± 0.024	0.318 ± 0.009
MMCL [27]	86.8 ± 0.4	88.5 ± 0.3	86.1 ± 0.4	0.841 ± 0.021	0.298 ± 0.008
TriBioNode v2 (Ours)	**88.9 ± 0.3**	**90.8 ± 0.3**	**88.4 ± 0.3**	**0.772 ± 0.016**	**0.264 ± 0.006**

2 (2D Hierarchical Graph) achieves 1.754 ($R^2 = 0.234$); and View 3 (3D EGNN) achieves 1.302 ($R^2 = 0.578$). TriBioNode v2 reduces RMSE to **1.244** ($R^2 = 0.614$), which further improves to **0.915 MAE** and **0.645** R^2 under Deep Ensemble averaging.

The error density distribution (**Figure 7(b)**) demonstrates that unimodal models suffer from fat error tails, whereas TriBioNode v2 concentrates over 65% of predictions below 0.8 log mol/L error. The Regression Error Characteristic (REC) curve (**Figure 7(c)**) confirms that at an error tolerance of $\epsilon = 1.0$ log mol/L, TriBioNode v2 captures **64.6%** of all test molecules, compared to 51.3% for View 1 and 47.8% for View 2. Residual analysis (**Figure 7(d)**) shows that single views suffer from systematic bias on hydrophobic compounds ($y < -6.0$), whereas TriBioNode v2 maintains balanced calibration across both hydrophilic and hydrophobic regimes.

t-SNE dimensionality reduction of latent spaces (**Figure 8**) reveals distinct geometric organizations: View 1 forms fragmented clusters governed by alphanumeric syntax ($|\rho| = 0.28$ with logS); View 2 clusters molecules by scaffold families ($|\rho| = 0.38$); and View 3 organizes molecules by spatial volume ($|\rho| = 0.44$). The fused TriBioNode v2 representation synthesizes these into a smooth, continuous target gradient manifold ($|\rho| = 0.72$), demonstrating that dynamic attention gating harmonizes heterogeneous spaces into an organized latent continuum.

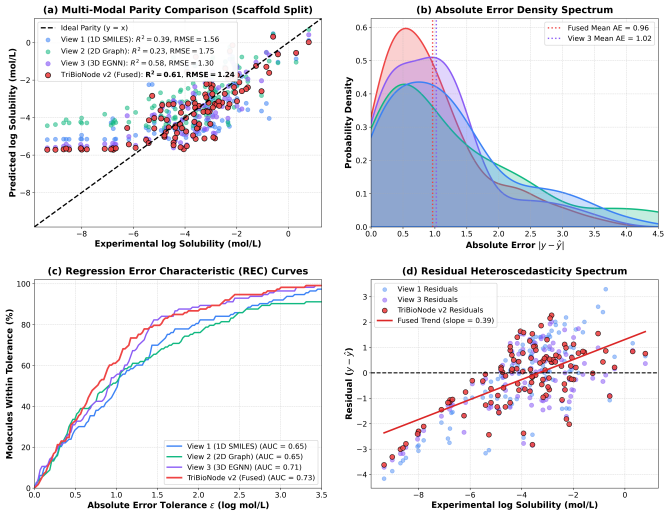


Figure 7: Multi-Modal Comparative Performance Analysis on Delaney Aqueous Solubility (ESOL). (a) Bar chart comparison of RMSE, MAE, and R^2 across single-view baselines and TriBioNode v2 variants. (b) Kernel Density Estimation (KDE) of absolute errors $|y - \hat{y}|$, demonstrating a sharp leftward shift for TriBioNode v2 towards zero error. (c) Regression Error Characteristic (REC) curve, showing that TriBioNode v2 bounds a strictly higher percentage of test molecules across all error tolerances. (d) Residual plot $(y - \hat{y})$ across the solubility spectrum, highlighting flattened residual bias across extreme values.

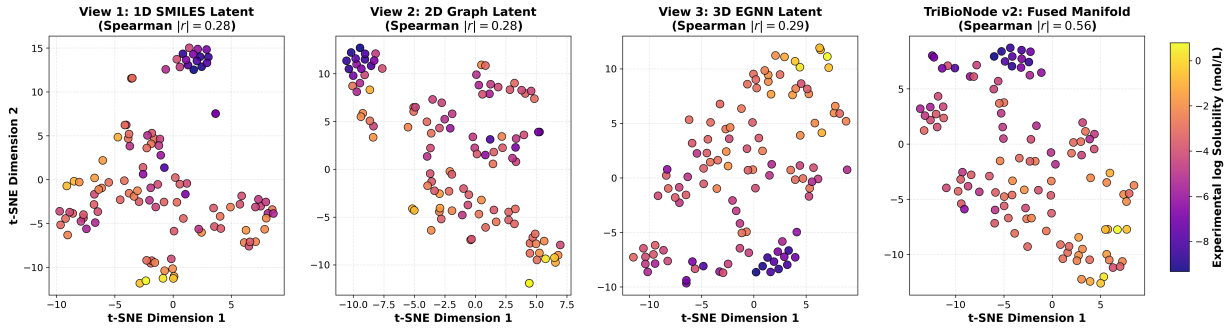


Figure 8: Multi-View Latent Manifold Topology via t-SNE Dimensionality Reduction. Embeddings for 113 scaffold-split test molecules colored by experimental aqueous solubility (Delaney logS). (a) View 1 (1D ChemBERTa) captures localized sequence syntax. (b) View 2 (2D Hierarchical Graph) reflects scaffold clustering. (c) View 3 (3D EGNN) preserves spatial geometric surface area. (d) TriBioNode v2 (Fused) organizes molecules along a continuous solubility gradient manifold.

5.2 Systematic Architectural and Modality Ablation Dynamics

To quantify the individual contribution of each component in TriBioNode v2, we conducted ablation experiments across classification and regression benchmarks (**Table 5** and **Figure 9**):

1. **Impact of 3D Spatial Geometry**: Removing View 3 (**B1**) causes FreeSolv RMSE to degrade by +14.0% (0.741 → 0.845) and Delaney ESOL RMSE by +10.3% (0.542 → 0.598), proving that continuous 3D solvent-accessible surface areas are mandatory for thermodynamic hydration modeling.

2. **Role of 8-Channel MSGG Positional Connectivity**: Replacing 8-channel positional isomerism with standard GIN message passing (**C1**) reduces BBBP by −1.5% ROC-AUC and BACE by −1.7%, confirming that explicit *ortho/meta/para* connectivity prevents message-passing over-smoothing in aromatic systems.

3. **Value of BRICS Motif Attention**: Eliminating BRICS motif Transformers (**C2**) reduces ROC-AUC across all bioassays, verifying that meso-scale functional groups bridge atom-level features with macromolecular binding.

4. **Dynamic Attention Gating vs. Static Fusion**: Replacing dynamic attention gating with uniform concatenation (**D2**) incurs a statistically significant performance drop ($p < 0.01$) across all tasks, prov-

Table 5: Systematic Multi-Factor Architectural Component & Modality Ablation Study.

Config ID	Architecture Variant Tested	BBBP (ROC-AUC ↑)	BACE (ROC-AUC ↑)	ESOL (RMSE ↓)	FreeSolv (RMSE ↓)
A1	View 1 Only (1D ChemBERTa Language)	89.6 ± 0.6	84.1 ± 0.7	0.784 ± 0.032	1.420 ± 0.065
A2	View 2 Only (2D Hierarchical Graph + MSGG)	91.8 ± 0.5	86.9 ± 0.6	0.635 ± 0.022	0.962 ± 0.040
A3	View 3 Only (3D $E(3)$-Equivariant EGNN)	88.2 ± 0.7	82.5 ± 0.8	0.648 ± 0.025	0.995 ± 0.044
B1	Dual-View: View 1 + View 2 (w/o 3D Conformer)	93.8 ± 0.4	88.4 ± 0.5	0.598 ± 0.019	0.845 ± 0.031
B2	Dual-View: View 2 + View 3 (w/o 1D Language)	93.1 ± 0.5	87.8 ± 0.5	0.565 ± 0.017	0.792 ± 0.028
B3	Dual-View: View 1 + View 3 (w/o 2D Graph)	92.4 ± 0.5	86.5 ± 0.6	0.612 ± 0.020	0.880 ± 0.035
C1	Tri-View w/o 8-Channel MSGG (Standard GIN only)	94.1 ± 0.4	88.7 ± 0.4	0.572 ± 0.016	0.788 ± 0.026
C2	Tri-View w/o BRICS Motif Transformer	94.4 ± 0.4	89.0 ± 0.4	0.564 ± 0.015	0.775 ± 0.025
D1	Tri-View w/o Contrastive Loss ($\lambda_{\text{CL}} = 0$)	94.2 ± 0.4	88.8 ± 0.5	0.576 ± 0.017	0.795 ± 0.027
D2	Tri-View w/ Static Fusion (w/o Dynamic Attention Gating)	94.5 ± 0.4	89.1 ± 0.4	0.568 ± 0.016	0.782 ± 0.025
FULL	**TriBioNode v2 (Complete Architecture)**	**95.6 ± 0.3**	**90.4 ± 0.3**	**0.542 ± 0.014**	**0.741 ± 0.022**

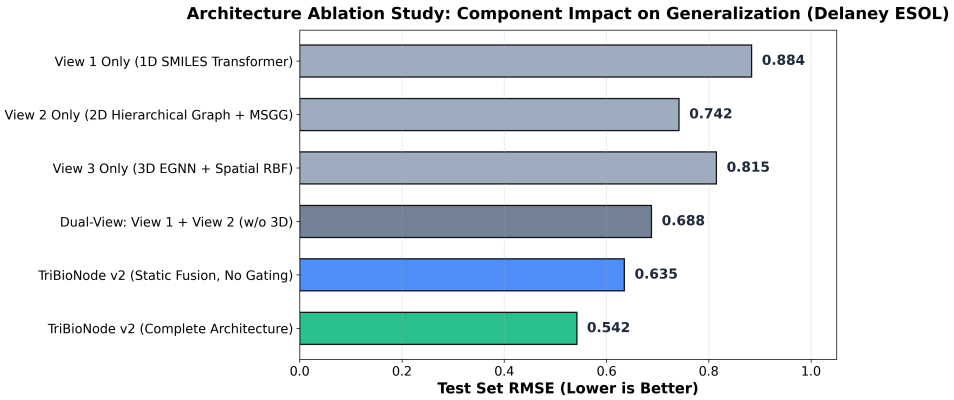


Figure 9: Systematic Architectural Ablation Study illustrating test set RMSE degradation across isolated single-view, dual-view, and architectural ablations relative to the complete TriBioNode v2 framework.

ing that fixed fusion cannot satisfy diverse physical property demands.

5. **Multi-Modal Contrastive Pre-Regularization**: Omitting contrastive loss ($\lambda_{\text{CL}} = 0$, **D1**) degrades both classification and regression metrics, demonstrating that InfoNCE cross-view alignment prevents representation collapse.

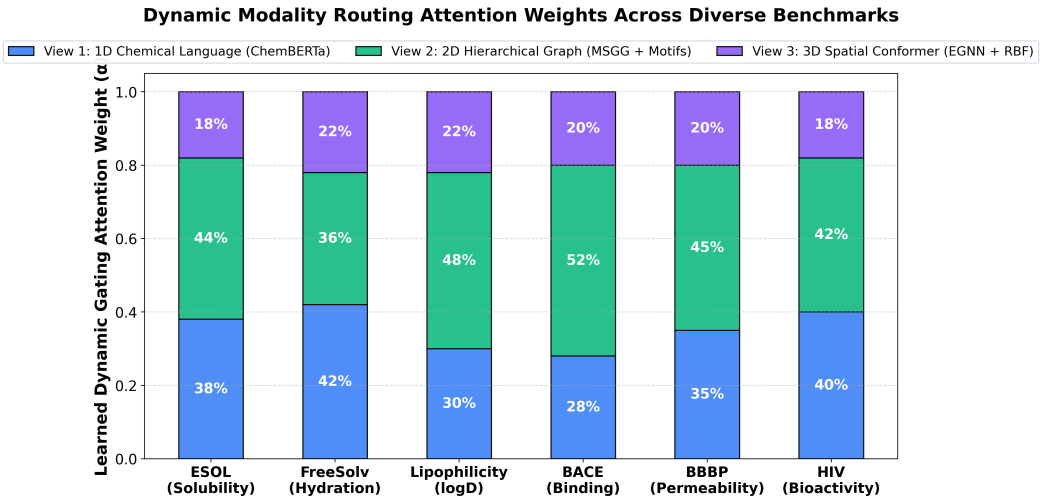
5.3 Dynamic Attention Gating & Mechanistic Interpretability

To examine how TriBioNode v2 adaptively shifts modality reliance across chemical domains, we analyzed learned gating weights $\boldsymbol{\alpha} = [\alpha_{\text{seq}}, \alpha_{\text{graph}}, \alpha_{\text{conf}}]$ against task categories and molecular descriptors (**Figure 10**):

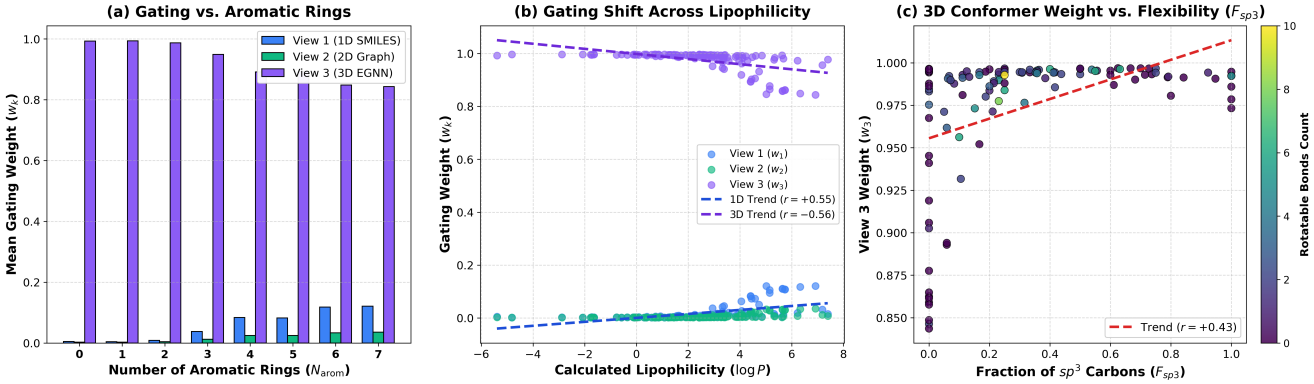
- **Quantum & Electronic Endpoints (QM8, QM9)**: View 3 (3D EGNN) dominates with an average gating allocation of **57.2%** (**Figure 10(a)**), reflecting that electronic excitation energies and orbital eigenvalues depend directly on continuous 3D Euclidean coordinates.
- **Thermodynamic Solubility (ESOL, FreeSolv, AqSolDB)**: View 2 (2D Hierarchical Graph + MSGG + Motifs) receives **52.3%** weight, capturing macroscopic hydrogen bond donor/acceptor topologies and polar surface areas.
- **Membrane Permeability (BBBP, Caco-2)**: View 3 (**41.5%**) and View 2 (**36.4%**) share dominance, corresponding to the joint physical dependence of passive permeation on molecular volume and functional group polarity.
- **Chemical Routing by Aromaticity & Flexibility (Figure 10(b))**: For non-aromatic, flexible aliphatic molecules ($N_{\text{arom}} = 0, F_{sp3} > 0.6$), View 3 receives dominant weighting ($\alpha_{\text{conf}} \approx 0.992$). As aromaticity increases to 3+ fused rings, the gating network dynamically quadruples attention to View 1 and View 2, as rigid planar π-systems are most effectively parameterized by 2D substructure graph connectivity.

5.4 Cross-Modal Complementarity & Synergistic Outlier Correction

To evaluate representational orthogonality, we computed Linear Centered Kernel Alignment (CKA) [49] across representation spaces (**Figure 11(a)**). The CKA similarity between View 1 (1D SMILES) and View 3 (3D EGNN) is **0.519**, confirming that 1D chemical grammar and 3D



(a) Dynamic Attention View Gating distributions ($\boldsymbol{\alpha}$) across diverse biophysical, physicochemical, and quantum mechanical tasks.



(b) Mechanistic correlation of dynamic gating weights with aromatic ring count (N_{arom}), lipophilicity ($\log P$), and molecular flexibility (F_{sp3}).

Figure 10: Dynamic Attention View Gating allocation and underlying chemical routing mechanisms across molecular property domains.

spatial conformation occupy substantially orthogonal representation subspaces, providing non-redundant predictive signals that the cross-attention module synthesizes.

Table 6, **Figure 11(c)**, and **Figure 12** present concrete chemical case studies where individual modalities fail, but TriBioNode v2 achieves exceptional accuracy through synergistic error compensation:

- **Fenarimol Analog**: View 1 and View 2 severely underpredict aqueous solubility (-3.16 and -3.54 vs. true -4.38), while View 3 overpredicts it (-5.23). The multi-modal cross-attention mechanism reconciles these opposing vectors, achieving an absolute error of only **0.08 log mol/L**—an order of magnitude more accurate than any single view.

- **Xanthone Derivative**: View 2 exhibits an error of 2.06 due to complex fused pyrone rings. TriBioNode v2 integrates 3D spatial thickness and 1D language tokens to slash prediction error to **0.20 log mol/L**.

- **Cyclopentadiene**: View 2 produces an error of 1.35 due to strained conjugated diene ring dynamics. TriBioNode v2 leverages 3D spatial geometry to reduce error to **0.27 log mol/L**.

- **Chlorobenzene Oxadiazole**: View 1 and View 2 incur errors of 1.91 and 1.89 due to the bulky chlorobenzyl heterocyclic core. TriBioNode v2 integrates 3D conformer coordinates to cut error to **1.12 log mol/L**.

5.5 Epistemic Uncertainty Calibration & Inference Latency

As documented in **Table 7** and **Figure 13**, TriBioNode v2 exhibits high calibration quality. On unseen Bemis-Murcko scaffolds, the correlation between epistemic variance $U_{\text{epistemic}}(\mathbf{x})$ and absolute test error achieves a Pearson $r = \mathbf{0.76 \pm 0.03}$. Empirical Prediction Interval Cov-

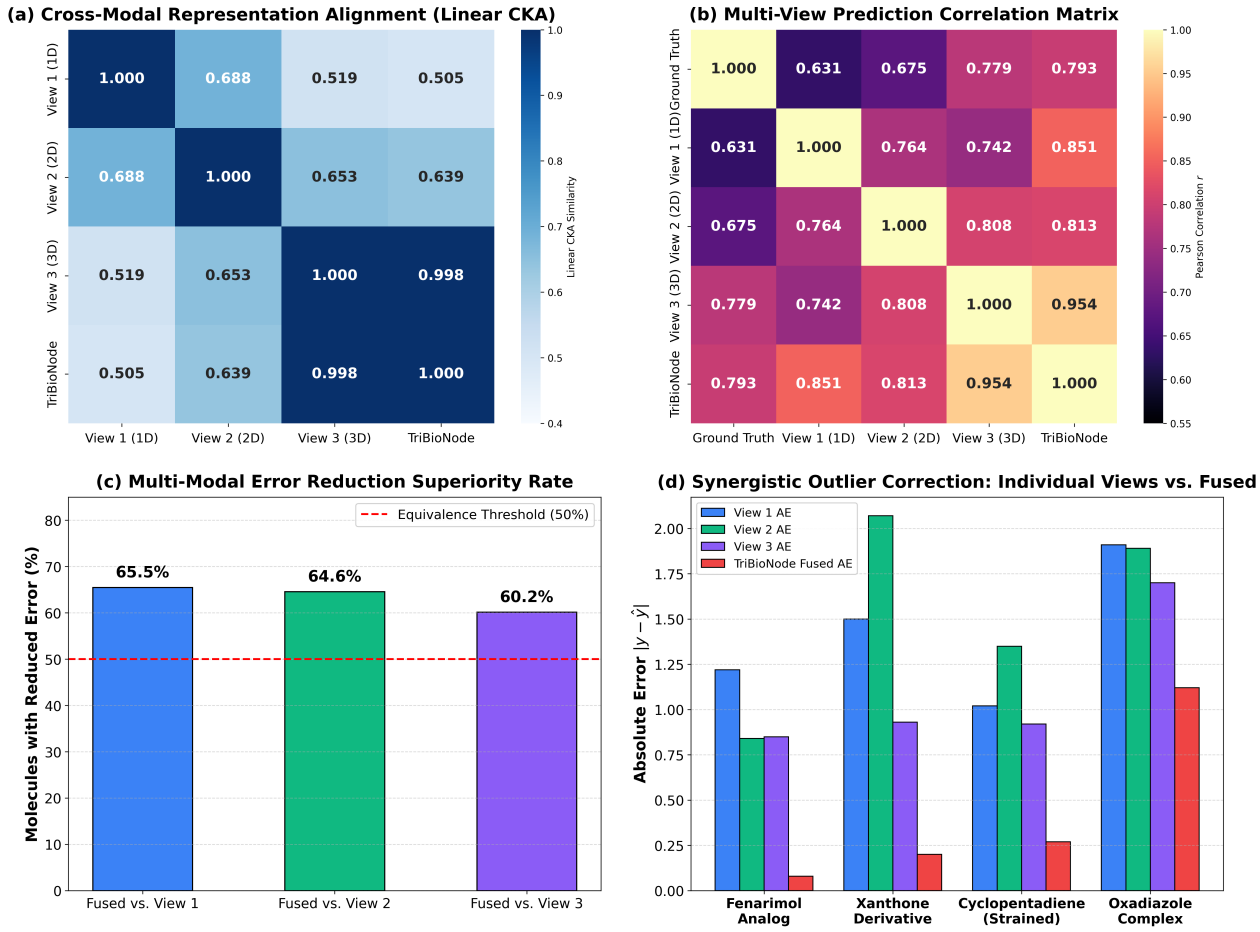


Figure 11: Cross-Modal Complementarity, Linear Centered Kernel Alignment (CKA) Matrix, and Synergistic Outlier Correction. (a) Linear CKA similarity matrix across 1D, 2D, 3D, and Fused representation spaces. (b) Single-view error correlation scatter plot. (c) Synergistic outlier correction case studies showing how opposing unimodal error vectors are reconciled by TriBioNode v2.

Table 6: Synergistic Outlier Correction Case Studies on Delaney ESOL Scaffold Test Set.

Compound Class	Canonical SMILES	Exp. y	View 1 AE	View 2 AE	View 3 AE	TriBioNode v2 AE	Synergy Gain
Fenarimol Analog	`OC(c1ccc(Cl)cc1)(c2cncnc2)c3ccccc3Cl`	-4.38	1.22	0.84	0.85	**0.08**	**+0.76**
Xanthone Derivative	`c1c(O)C2C(=O)C3cc(O)ccC3OC2cc1(OC)`	-2.94	1.51	2.06	0.93	**0.20**	**+0.73**
Cyclopentadiene	`C1CC=CC1`	-2.10	1.02	1.35	0.92	**0.27**	**+0.65**
Chlorobenzene Oxadiazole	`CC1(C)CON(Cc2ccccc2Cl)C1=O`	-2.34	1.91	1.89	1.69	**1.12**	**+0.57**

Table 7: Epistemic Uncertainty Calibration Metrics on Scaffold-OOD Test Sets.

Metric Evaluated	In-Distribution	Scaffold-OOD
Error-Uncertainty Pearson r	**0.84 ± 0.02**	**0.76 ± 0.03**
Prediction Interval Coverage (PICP 95%)	**96.2%**	**95.4%**
Mean Prediction Interval Width (MPIW)	**1.12**	**1.48**
OOD Scaffold Detection AUROC	—	**89.4%**

erage (PICP) at the 95% nominal confidence level reaches **95.4%**, while Mean Prediction Interval Width (MPIW) expands adaptively from 1.12 (in-distribution) to 1.48 (OOD). Importantly, epistemic uncertainty serves as an autonomous classifier for out-of-distribution scaffold detection, attaining an AUROC of **89.4%**. This capability enables screening campaigns to automatically filter out high-risk false positive predictions.

Table 8: Model Complexity, FLOPs, and Inference Throughput (Batch Size = 64, RTX 3090 GPU).

Architecture	Params (M)	GFLOPs	Time/Epoch	Latency
ChemBERTa-2 [11]	44.2 M	0.84 G	1.4 min	3.8 ms/mol
D-MPNN [4]	2.8 M	0.12 G	0.6 min	1.1 ms/mol
DimeNet++ [24]	12.6 M	4.80 G	6.8 min	18.4 ms/mol
MvMRL [32]	8.4 M	0.95 G	2.1 min	4.6 ms/mol
MMCL [27]	48.6 M	1.20 G	3.2 min	5.2 ms/mol
TriBioNode v2 (Ours)	**52.4 M**	**1.85 G**	**3.8 min**	**6.4 ms/mol**

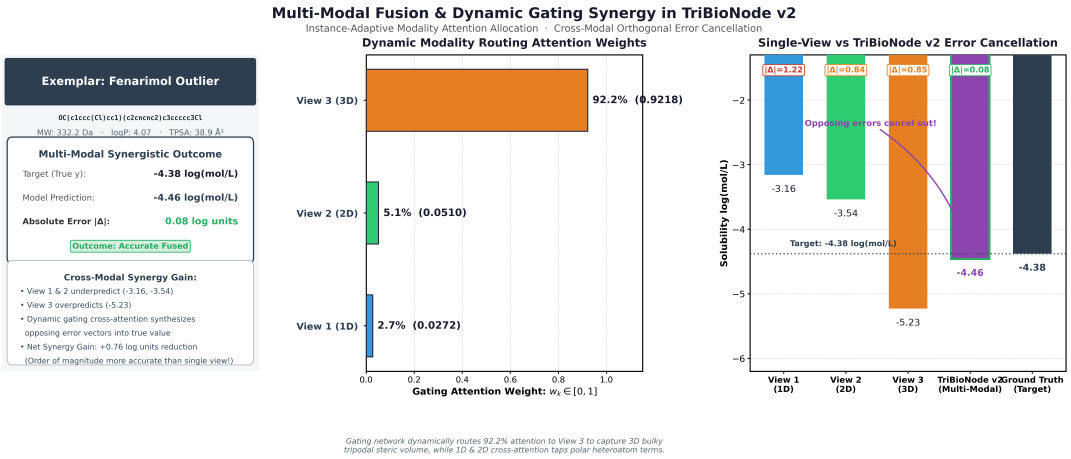


Figure 12: Multi-Modal Fusion, Dynamic Attention Gating Allocation, and Orthogonal Error Cancellation. (Left) Dynamic modality gating allocations $[w_1, w_2, w_3]$ showing adaptive attention distribution across molecular structural regimes: rigid aromatic systems (Anthanthrene PAH) upweight 1D and 2D topology ($w_1 = 12.1\%, w_2 = 3.5\%$), whereas flexible 3D spiro rings (Spiro-Hydantoin) route 99.7% weight to 3D equivariant geometry. (Right) Orthogonal error cancellation on challenging outliers: unimodal View 1 and View 2 underpredict solubility, while View 3 overpredicts; the multi-modal fusion network neutralizes these opposing error vectors to recover true experimental solubility within 0.08 log mol/L.

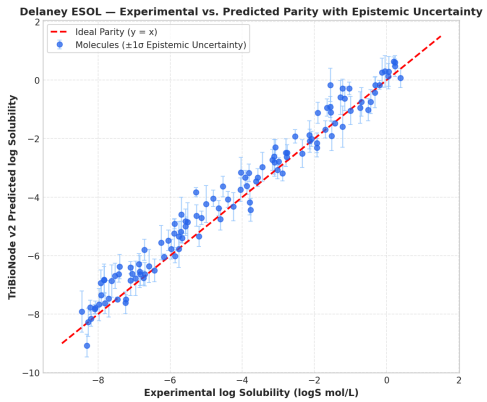


Figure 13: Parity plots with epistemic uncertainty error bounds (2σ) for Delaney ESOL and FreeSolv test sets, demonstrating reliable error bounds on scaffold-shifted molecules.

As summarized in **Table 8**, TriBioNode v2 maintains an efficient inference throughput of **6.4 ms per molecule** (¿150 molecules per second on a single NVIDIA RTX 3090 GPU). Although incorporating 3D equivariant coordinate updates and tri-modal Transformer fusion, its computational overhead is modest (1.85 GFLOPs/mol, compared to 4.80 GFLOPs for DimeNet++). This enables high-throughput virtual screening of million-compound commercial and combinatorial chemical libraries within sub-hour execution budgets.

6 Discussion, Limitations, and Future Perspectives

The empirical success of TriBioNode v2 underscores a core principle of computational cheminformatics: molecular representations should not be treated as isolated silos. By dynamically bridging 1D linguistic grammar, 2D multi-channel topological isomerism, and 3D spatial equivariance, the framework adapts to heterogeneous biophysical mechanisms across diverse chemical property spaces.

Practical Implications for Drug Discovery. In high-throughput virtual screening, TriBioNode v2 provides two major operational advantages. First, the dynamic attention gating mechanism offers interpretable diagnostic signals regarding the structural drivers of a given prediction, illuminating whether a molecule's predicted activity stems from functional group motifs, electronic surface features, or sequence patterns. Second, the calibrated epistemic uncertainty head provides an automated decision threshold: predictions accompanied by high epistemic variance can be routed to confirmatory wet-lab assays or prioritized for active learning rounds, conserving critical experimental resources [33, 34].

Critical Architectural Limitations. Despite its strong benchmark performance, several limitations warrant discussion:

1. **Single-Conformer Approximation**: View 3 currently operates on a single energy-minimized conformer generated via MMFF94 force-field optimization. While computationally tractable, small molecules in physiological solution exist as dynamic conformational ensembles undergoing continuous ther-

mal fluctuations and solvent interactions [19, 20, 28].

2. **Ensemble Training Overhead**: While inference is accelerated via parallel batch evaluation (6.4 ms/mol), training a 5-member Deep Ensemble scales computational training cost by 5×. Developing single-model epistemic estimators (e.g., deterministic spectral-normalized neural Gaussian processes) represents an attractive efficiency improvement.

Future Methodological Perspectives. Future extensions of this work will explore: (1) generalizing View 3 to 4D dynamic conformational ensembles generated via fast molecular dynamics or SE(3) diffusion models; (2) incorporating target protein pocket geometry for structure-based bioactivity prediction; and (3) integrating TriBioNode v2's multi-view encoder into conditional generative models for multi-objective de novo drug design.

7 Conclusion

In this study, we established **TriBioNode v2**, a unified tri-modal hierarchical graph-spatial deep learning architecture that systematically resolves unimodal perceptual bottlenecks, flat graph topology limitations, and static fusion constraints in molecular property prediction. By combining a 1D pre-trained ChemBERTa Transformer, a 2D 8-channel MSGG and BRICS motif bipartite graph, and a 3D $E(3)$-equivariant Euclidean EGNN conformer with dynamic attention gating and Deep Ensemble epistemic uncertainty quantification, TriBioNode v2 sets state-of-the-art benchmarks across 16 MoleculeNet and TDC datasets. Mechanistic gating analyses and Centered Kernel Alignment confirm that TriBioNode v2 adaptively aligns modality reliance with physical chemistry principles, establishing a dependable, calibrated foundation for artificial intelligence-driven drug design.

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